1 Experimental Implementation using SDR and MATLAB

1.a Experimental Layout and Network Topology

In the (following) table we can see the parameters of the bpsk layout

Table 1: Key SDR Hardware and Software Parameters for the OAC Implementation

Parameter	**Value**	**Description**
Hardware Configurations		
Center Frequency (f_c)	2.4 GHz	ISM band, selected to minimize external interference.
Sample Rate (f_s)	32 kHz	Defined baseband rate for the SDR sink/source blocks.
TX / RX Gain	TBD	Adjusted during runtime to prevent clipping while maintaining sufficient SNR.
Antenna Configuration	SISO	Single-Input Single-Output for initial baseline testing.
Software & Baseband Processing		
Modulation Scheme	BPSK	Binary Phase Shift Keying (Symbol mapping: $-1, 1$).
Samples per Symbol (sps)	6	Determines the interpolation/decimation rate for the signal.
Pulse Shaping Filter	RRC	Root Raised Cosine filter applied to mitigate Inter-Symbol Interference (ISI).
Roll-off Factor (α)	0.3	Standard excess bandwidth factor defining the filter's frequency transition band.

- *Layout Description*: Presentation of the network structure (e.g., a single-cell OAC scenario or an ad-hoc network) using Software Defined Radio (SDR) to represent Edge Devices (EDs) and a Fusion Center (FC)[1].
- *Hardware and Software Tools*: Specific mention of the SDR and the use of MATLAB for waveform generation, digital signal processing, control loops, and simulation[1], [2], [3], [4].

1.b Design and Implementation of the OAC Protocol

1.b.i Selection of the OAC Scheme for Demonstration

Choice of Target Function (e.g., Arithmetic Mean or Majority Vote): Justification based on practical implementation constraints of SDR[1].

1.b.ii Synchronization Mechanisms

Detailed methodology for achieving coarse block synchronization or mitigating synchronization errors (TO/CFO) in the SDR/MATLAB environment (e.g., based on trigger signals or custom control loops)[1], [3], [4].

1.b.iii Pre-processing and Channel Compensation Implementation

Implementation of the chosen modulation/encoding scheme and any necessary pre-equalization (e.g., Channel Inversion precoding if a coherent scheme is selected, or non-coherent coding if robustness is prioritized)[1], [4].

1.c Measurement and Data Collection Methodology

Procedure for varying network parameters (e.g., number of transmitting devices, transmit SNR)[5].

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